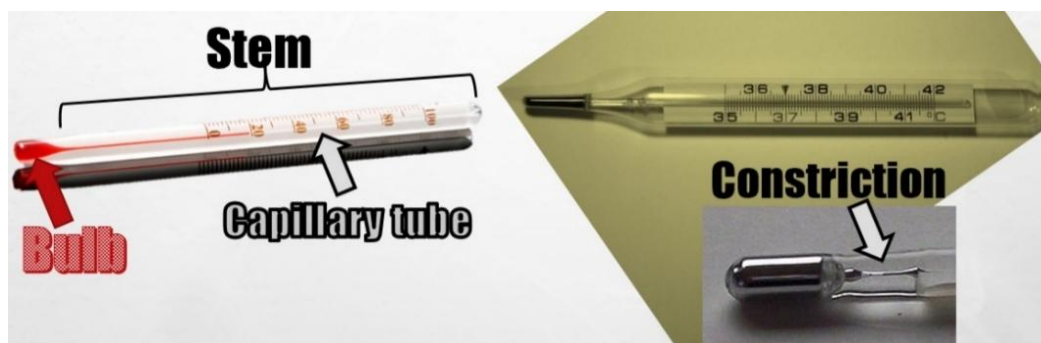


Temperature and thermometer

- Temperature is a measure of the **degree of hotness** of an object. **Thermometer** is used to measure the temperature of objects.
- Degree Celsius ($^{\circ}\text{C}$) is a commonly used unit of temperature.

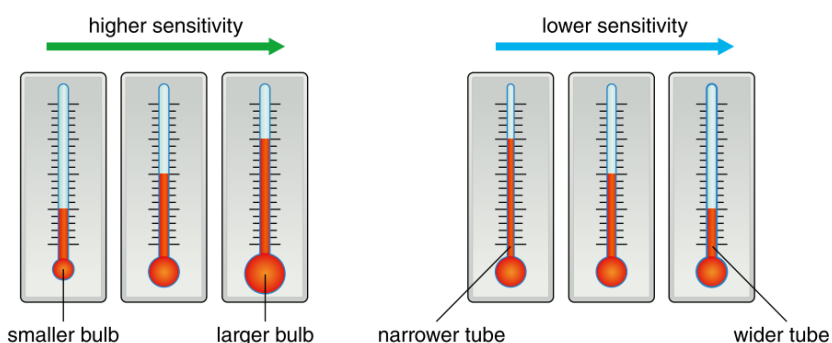


- Fixed point are the **reproducible** temperatures in the laboratory
 0°C : Ice point (lower fixed point)
 Temperature of pure melting ice at normal atmospheric pressure

 100°C : Steam point (upper fixed point)
 Temperature of steam over pure boiling water at normal atmospheric pressure

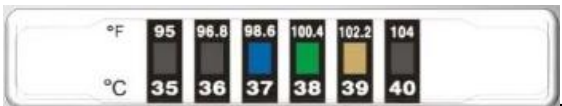
- A liquid-in-glass thermometer is **more sensitive** if the length of liquid column increases more for the same temperature rise. Thermometers with the following features are more sensitive:

- (1) **Narrower** capillary tube
- (2) **Larger** blub
- (3) Containing liquid that **expands more** when temperature increases



Sensitivity $\uparrow$	Narrower Capillary tube
Response $\uparrow$	Thinner Glass wall
Range $\uparrow$	Smaller Bulb

Type of thermometers

Thermometer	Temperature dependent property
Liquid in glass thermometer 	Volume of liquid (Volume increase as temperature increase)
Infra-red thermometer 	Intensity of infra-red radiation (Intensity increase as temperature increase)
Rotary thermometer 	Curvature of bimetallic strip High expansion metal  Low expansion metal  (When heated , the strip bends because metals expand with different amounts)
Thermistor thermometer 	Electrical resistance of thermistor (Resistance decrease as temperature increase)
Resistance thermometer 	Electrical resistance of metal (Resistance increase as temperature increase)
Liquid crystal thermometer 	Colour of liquid crystal (Colour changes as temperature changes)

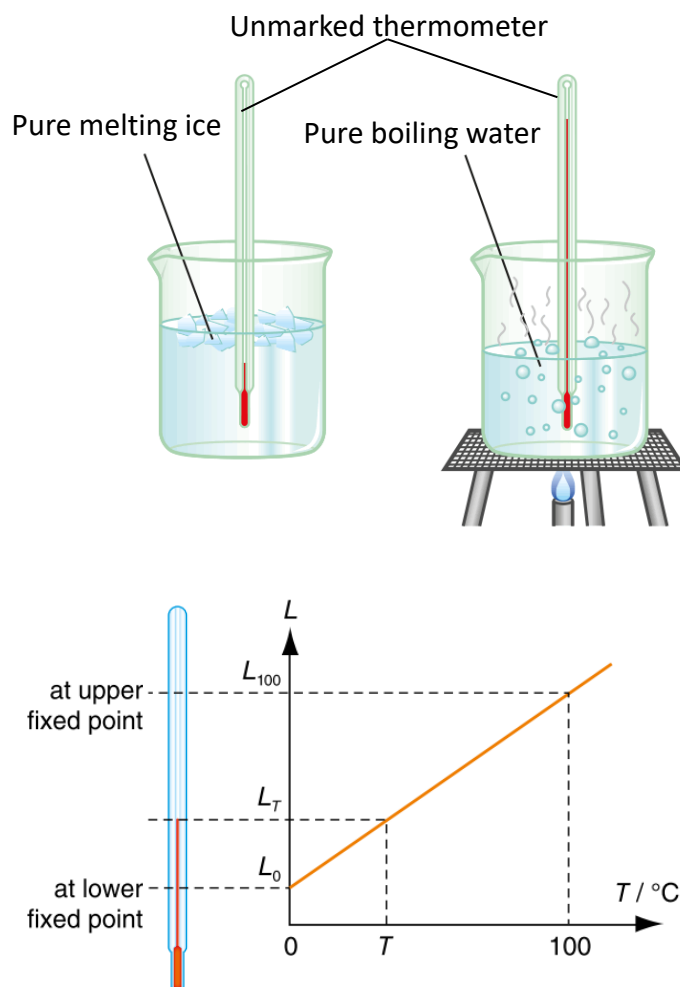
Comparison between thermometers

Mercury in glass thermometer	Alcohol in glass thermometer
	
Can measure high temperature	Can measure low temperature
Quick response	Slow response
Toxic	Non-toxic

Calibration of unmarked thermometer

- Put the thermometer in **pure melting ice** (at lower fixed point, 0°C) and then in **pure boiling water** (at upper fixed point, 100°C).
- The liquid level in each case is marked.
- Divided the length between two markings into **100 equal divisions**. Each division represent 1°C .

Remarks: In the above calibration, we assume the length of the liquid column **varies linearly** with temperature. However, the temperature-dependent properties of other types of thermometers may not vary linearly with temperature.



1. State which type of thermometer you would use to measure the temperature of very hot water.
 - A. Clinical thermometer
 - B. Alcohol-in-glass thermometer
 - C. Mercury-in-glass thermometer
 - D. Resistance thermometer C
2. State which type of thermometer you would use to measure the temperature of 'dry-ice' at about -80°C .
 - A. Clinical thermometer
 - B. Alcohol-in-glass thermometer
 - C. Mercury-in-glass thermometer B
 - D. Resistance thermometer
3. The freezing point and the boiling point of sea water is not suitable for use as fixed points because D
 - A. sea water is salty.
 - B. sea water has a higher density than pure water.
 - C. sea water is corrosive.
 - D. different samples of sea water may have different freezing points and boiling points.
4. The temperatures of pure melting ice and pure boiling water at standard atmospheric pressure are chosen as the fixed points of the Celsius scale. This is because D
 - (1) the two temperatures can be reproduced easily.
 - (2) pure ice always melt at 0°C under standard atmospheric pressure.
 - (3) pure water always boil at 100°C under standard atmospheric pressure.
 - A. (1) only
 - B. (1) and (2) only
 - C. (2) and (3) only
 - D. (1), (2) and (3)
5. Which of the following about the Celsius scale is/are correct? D
 - I. The lower fixed point is 0°C and the upper fixed point is 100°C .
 - II. Temperature measured is between 0°C and 100°C only.
 - III. Temperature between 0°C and 100°C are divided into 100 equal divisions.
 - A. I only
 - B. II only
 - C. III only
 - D. I and III only
6. If the mercury in a liquid-in-glass thermometer is replaced by alcohol, which of the following statements is/are correct? D
 - (1) The thermometer can measure a higher temperature.
 - (2) The thermometer becomes less sensitive to temperature change.
 - (3) The thermometer will be safer to use.
 - A (2) only
 - B (3) only
 - C (1) and (2) only
 - D (2) and (3) only

temperature range that can be measured by:

clinical thermometer —
around 37°C , our body's temperature

alcohol-in-glass thermometer —
around -115°C to 78°C

mercury-in-glass thermometer —
around -35°C to 357°C

resistance thermometer —
around -200°C to $1\,000^{\circ}\text{C}$

7. Why should the stem of a mercury-in-glass thermometer be made thin?

- A. Save the use of mercury.
- B. It is lighter in weight.
- C. It can be more sensitive to temperature change.
- D. It can measure higher temperature.

C

Temperature increases

⇒ volume of mercury expands

If stem is thin ⇒ length increase is more

⇒ easier to observe

8. Which of the following can increase the sensitivity (length increase per °C of temperature rise) of a mercury-in-glass thermometer?

- (i) make the glass tube longer
- (ii) make the glass tube narrower
- (iii) make the bulb larger

- A. (i) only
- B. (ii) only
- C. (i) and (ii) only
- D. (ii) and (iii) only

D

9. A clinical thermometer shows a patient's body temperature even after it has been taken out from his mouth. This is because

- A the capillary tube has a very narrow bore.
- B it has a constriction near the bulb.
- C the density of mercury is very high.
- D the stem of the thermometer contracts.

(B)

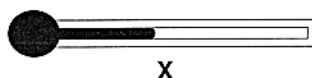
10. Melanie wants to make a liquid-in-glass thermometer which can read a wider range of temperature. Which of the following ways will help?

- 1 Use a smaller bulb.
- 2 Use a narrower capillary tube.
- 3 Use mercury instead of alcohol.

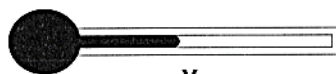
- A 1 and 2 only
- B 1 and 3 only
- C 2 and 3 only
- D 1, 2 and 3

(B)

11. The figures below shows three mercury thermometers, **X**, **Y** and **Z**, which are identical in every aspect except **Y** has a bigger bulb and **Z** has a smaller bore capillary tube.



X



Y

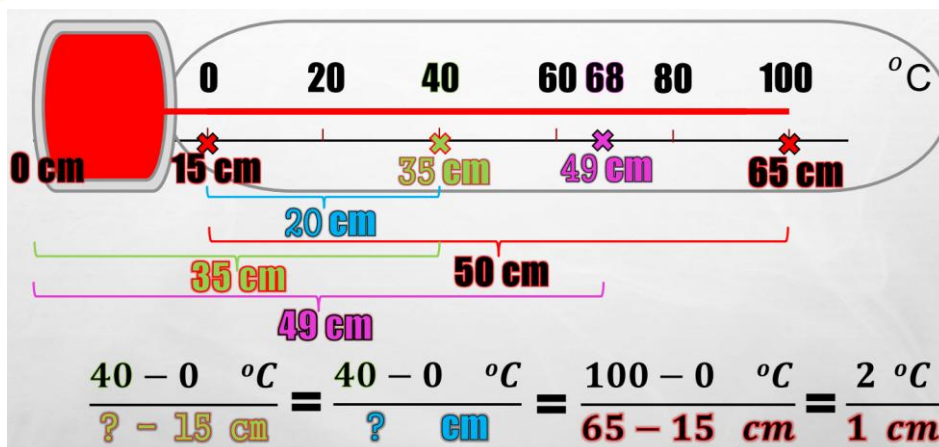


Z

Which of the following correctly describes sensitivity and range of the thermometer?

	Sensitivity	Range
A	Z > X	Y > Z
B	Z > Y	X > Z
C	Z > X	X > Y
D	Y > Z	X > Z

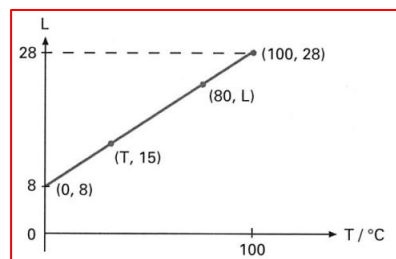
(C)



(For question 12 and 13) refer to the following:

The length of mercury column when a mercury in glass thermometer is put in 0°C and 100°C are 8 cm and 28 cm respectively

12. Find the temperature of the object it measures when the length of mercury column is 15 cm



Find the slope

$$\frac{28 - 8}{100 - 0} = \frac{15 - 8}{T - 0} \Rightarrow T = 35^{\circ}\text{C}$$

13. Find the length of mercury column if it is used to measure an object of temperature 80°C

$$\frac{28 - 8}{100 - 0} = \frac{L - 8}{80 - 0} \Rightarrow L = 24 \text{ cm}$$

14. When a mercury-in-glass thermometer is put in pure melting ice and then in pure boiling water, the lengths of the mercury thread are 3 cm and 19 cm respectively. When the thermometer is put in a liquid, the length of the mercury thread is 9 cm. What is the temperature of the liquid?

$$\begin{aligned} \frac{T}{100} &= \frac{9 - 3}{19 - 3} \\ \Rightarrow T &= \frac{6}{16} \times 100 = 37.5^{\circ}\text{C} \end{aligned}$$

15. The resistance of an alloy wire is 25Ω when it is placed in a cup of melting ice. At room temperature of 28°C , the resistance is 38Ω . If the wire is placed in boiling water, what will be the resistance?

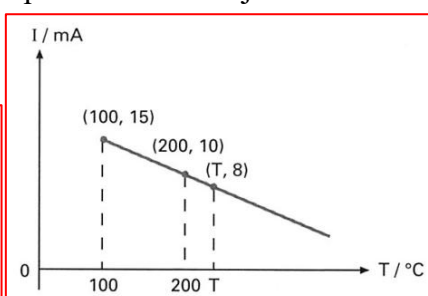
$$71\Omega$$

16. When the metal wire is put into 100°C water, ammeter reads 15 mA. When the metal wire is put into a liquid of 200°C , ammeter reads 10 mA. If it is used to measure the temperature of an object and the ammeter reads 8 mA, find the temperature of the object.

Find the slope

$$\frac{15 - 10}{100 - 200} = \frac{10 - 8}{200 - T}$$

$$T = 240^\circ\text{C}$$



17. A faulty thermometer has a wrong calibration. It reads 5°C and 85°C when the actual temperatures are 0°C and 100°C respectively. If the faulty thermometer reads 45°C , what is the real temperature?

$$B \quad \frac{T}{100} = \frac{45 - 5}{85 - 5}$$

$$\Rightarrow T = \frac{40}{80} \times 100 = 50^\circ\text{C}$$

(For question 18 and 19)

A certain thermometer is found to read -10°C in pure melting ice and 110°C in steam. Assume this faulty thermometer has a uniform stem and the divisions are of equal length.

18. What is the reading of thermometer when the actual temperature is 40°C

- A. 50°C
B. 48°C
C. 42°C
D. 38°C

$$\frac{40 - 0}{100 - 0} = \frac{x - (-10)}{110 - (-10)}$$

$$x = 38$$

D

19. What is the actual temperature when the reading of thermometer is 40°C

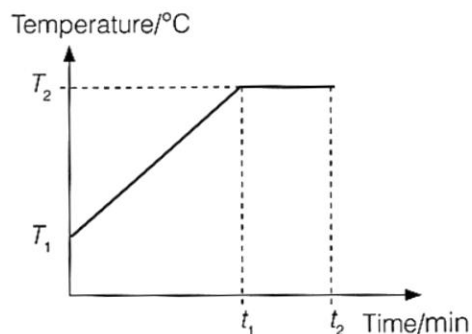
- A. 30°C
B. 38°C
C. 42°C
D. 58°C

$$\frac{40 - (-10)}{110 - (-10)} = \frac{x - 0}{100}$$

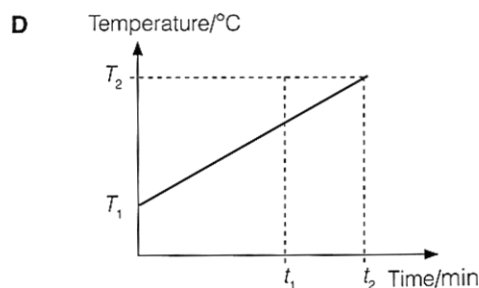
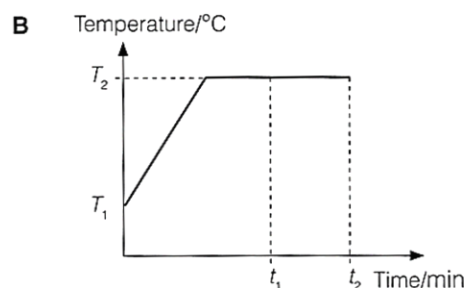
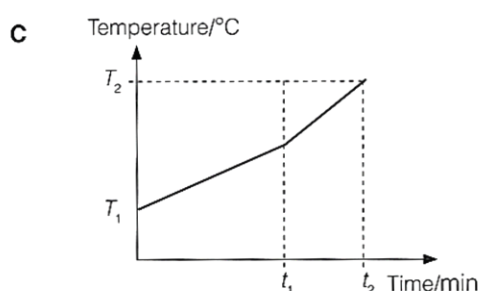
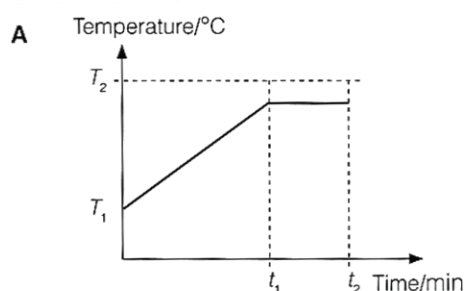
$$x = 41.7 \approx 42$$

C

20. Yong Hui heats up a bowl of water in an oven and discovers that the temperature-time graph is as shown below.

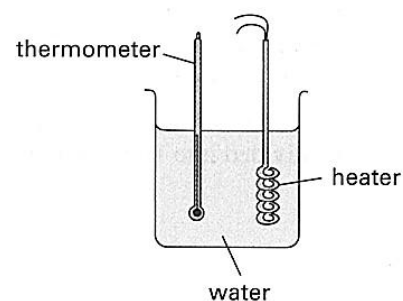
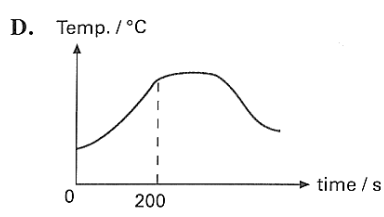
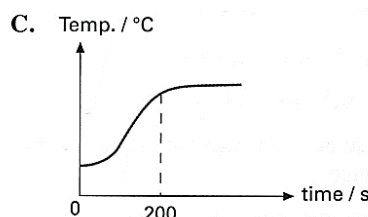
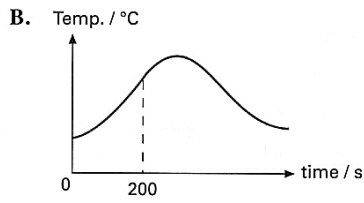
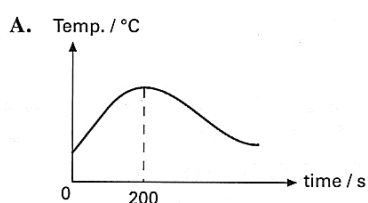


Which of the following temperature-time graphs should be obtained when she uses the same amount of water in a more powerful heater?



B

21. In the figure, a heater is used to heat the water for 200 s and then switched off. Which of the following temperature-time graphs of water is correct?



When the heater is switched off, it is still very hot (temperature is still higher than the liquid), so heat is still given out
⇒ temperature of liquid still rises for a while and then falls

B

22. A student calibrates an unmarked liquid-in-glass thermometer on the Celsius scale. She first put the thermometer in pure melting ice (Fig a) and then in pure boiling water (Fig b). The liquid level in each case is marked.



Figure a

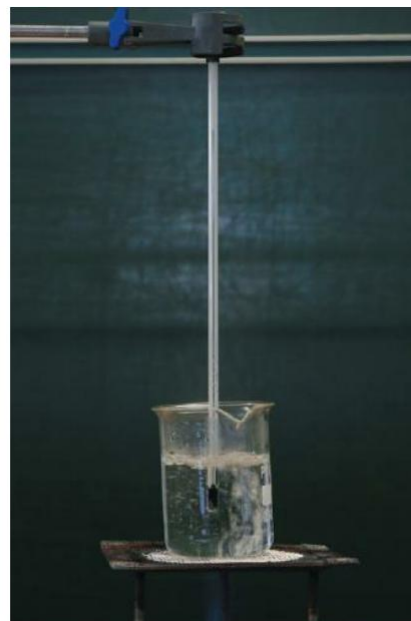


Figure b

The lengths of the liquid column of the thermometer in pure melting ice and pure boiling water are given in Figure c.

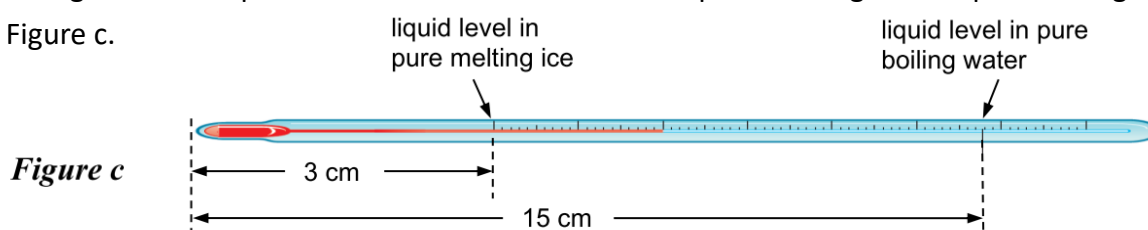
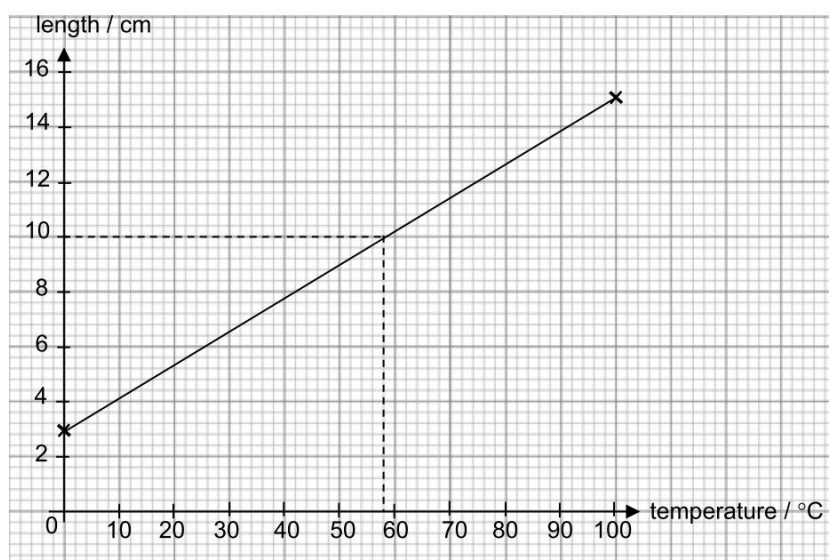


Figure c

- (a) Draw the calibration graph of the thermometer on graph paper. (3 marks)
- (b) The student puts the thermometer into a cup of hot tea. The length of the liquid column is 10 cm. What is the temperature of the hot tea as read from the graph in (a)? (1 mark)

(a)



(Correct labelled axes)
(Correct data points)
(Correct graph)

23. When a resistance thermometer is put in pure melting ice and pure boiling water, the resistances are $100\ \Omega$ and $140\ \Omega$ respectively. Assume resistance R varies linearly with temperature T .
- Draw a calibration graph in Figure b.
 - What temperature does the thermometer read when the resistance is $124\ \Omega$?
 - What is the resistance if the thermometer is in a room at $30\ ^\circ\text{C}$?

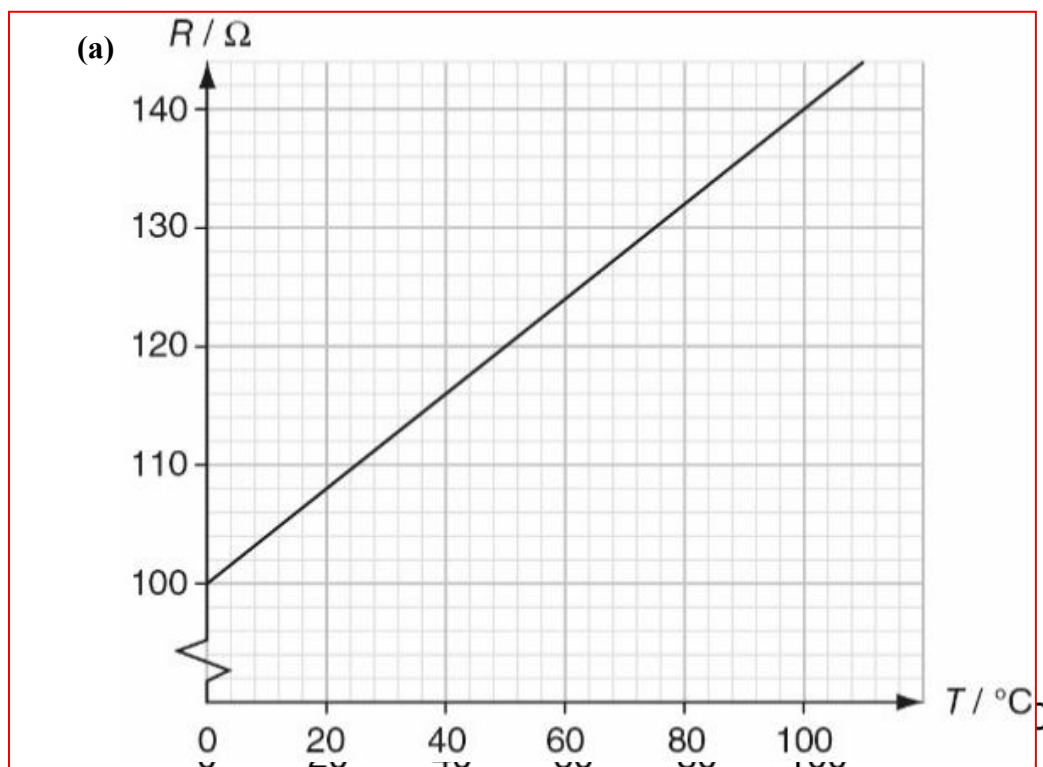


Fig b

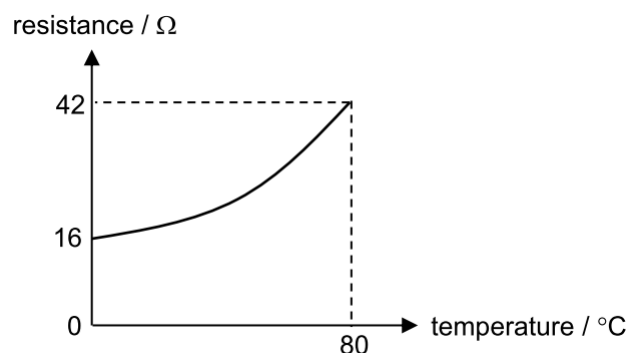
- (b) $60\ ^\circ\text{C}$
(c) $112\ \Omega$

24. A resistance thermometer is used to measure temperature. The following shows the resistance of the thermometer at 0 °C and 80 °C respectively.

Temperature / °C	Resistance / Ω
0	16
80	42

When the thermometer is put into a glass of water, its resistance is 31 Ω .

- (a) Assume the resistance of the thermometer increases linearly with temperature. Find the temperature of the water. (2 marks)
- (b) The figure shows how the resistance of the thermometer changes with temperature.
- (i) Is the actual temperature of the water higher or lower than the value found in (a)? (1 mark)
- (ii) Is the thermometer more sensitive at low temperatures or high temperatures? Explain briefly. (2 marks)



- (a) By proportion,

$$\frac{R - R_0}{R_{80} - R_0} = \frac{T - 0}{80 - 0}$$

$$\frac{31 - 16}{42 - 16} = \frac{T}{80}$$

$$T = 46.2 \text{ } ^\circ\text{C}$$

- (b) (i) Higher

- (ii) At high temperatures, the thermometer is more sensitive because the resistance changes more significantly for the same change in temperature.

1M

1A

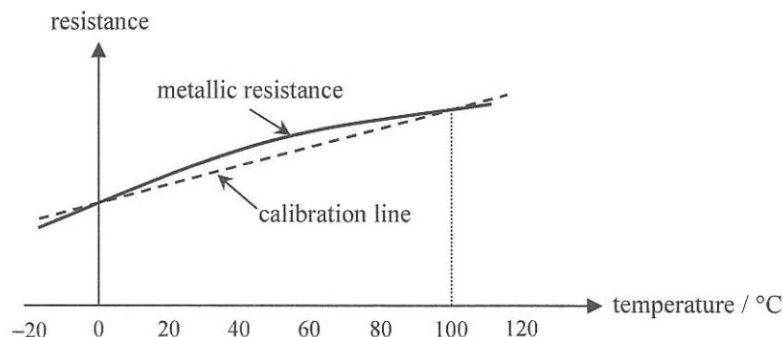
1A

1A

1A

25.

The solid curve in the Figure below shows how the resistance of a metallic resistance thermometer varies with temperature. This thermometer is calibrated at standard atmospheric pressure for the melting point of ice and the steam point of boiling water. The dotted calibration line represents how the resistance of the thermometer varies with temperature if a linear resistance-temperature relationship is assumed. The deviation of the curve from linearity has been exaggerated in the figure.



- (a) (i) Using the resistances at the calibration points tabulated below, calculate the expected resistance at 60°C if the resistance varies linearly with temperature. (2 marks)

temperature / °C	resistance / Ω
0	102.00
100	140.51

- (ii) Now if the resistance of the resistance thermometer is the value found in (a) (i), is the actual temperature higher than, lower than or equal to 60°C? (1 mark)

(a) (i) $\frac{R - R_0}{R_{100} - R_0} = \frac{\theta - 0}{100 - 0}$ [1]

$\therefore \frac{R - 102.00}{140.51 - 102.00} = \frac{60}{100}$

$\therefore R = 125.106 \Omega$ < accepted 125.11 Ω > < 125 Ω and 125.1 Ω not accepted > [1]

(ii) Actual temperature is lower than 60°C. [1]